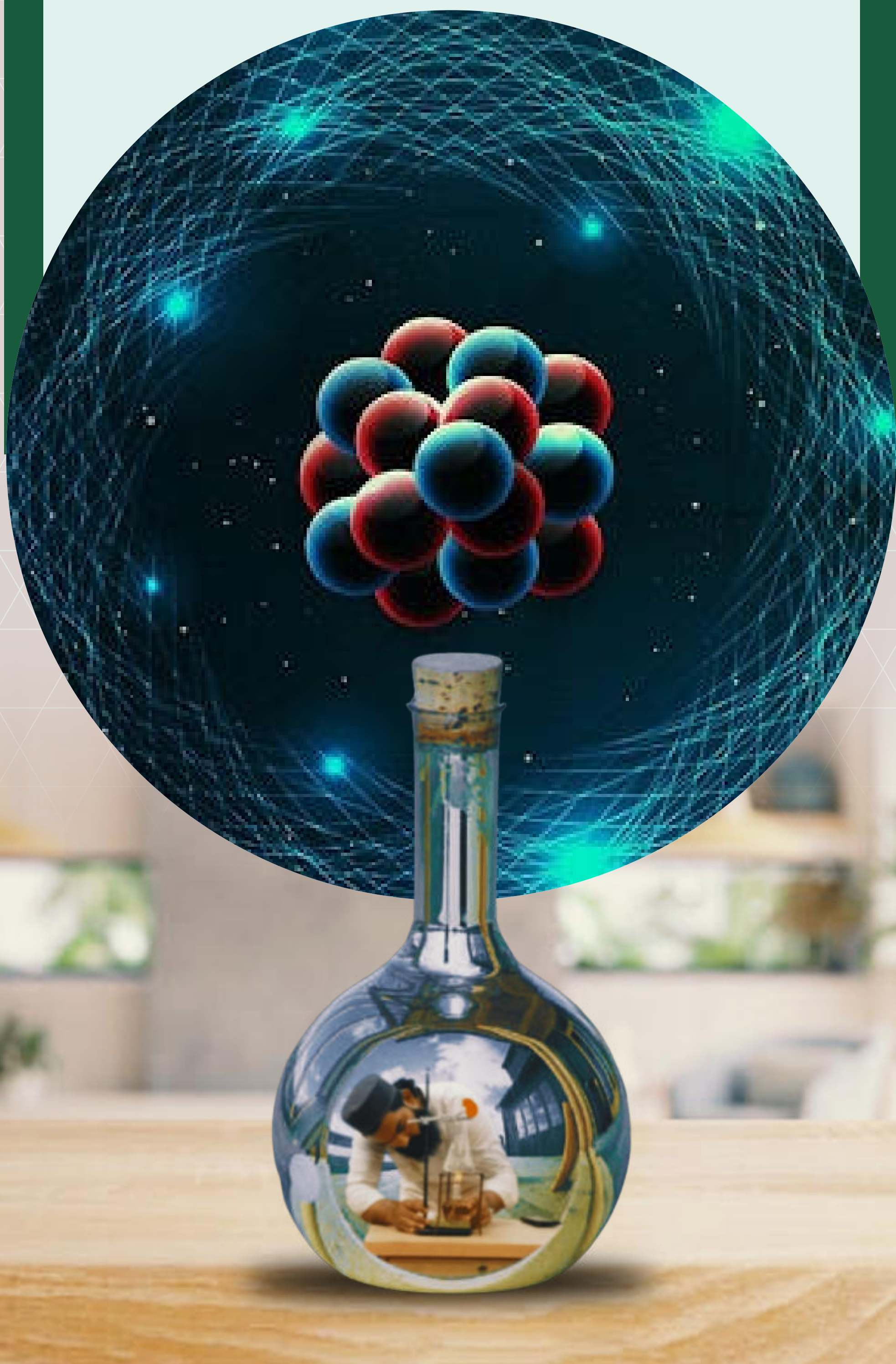
UNIT 05

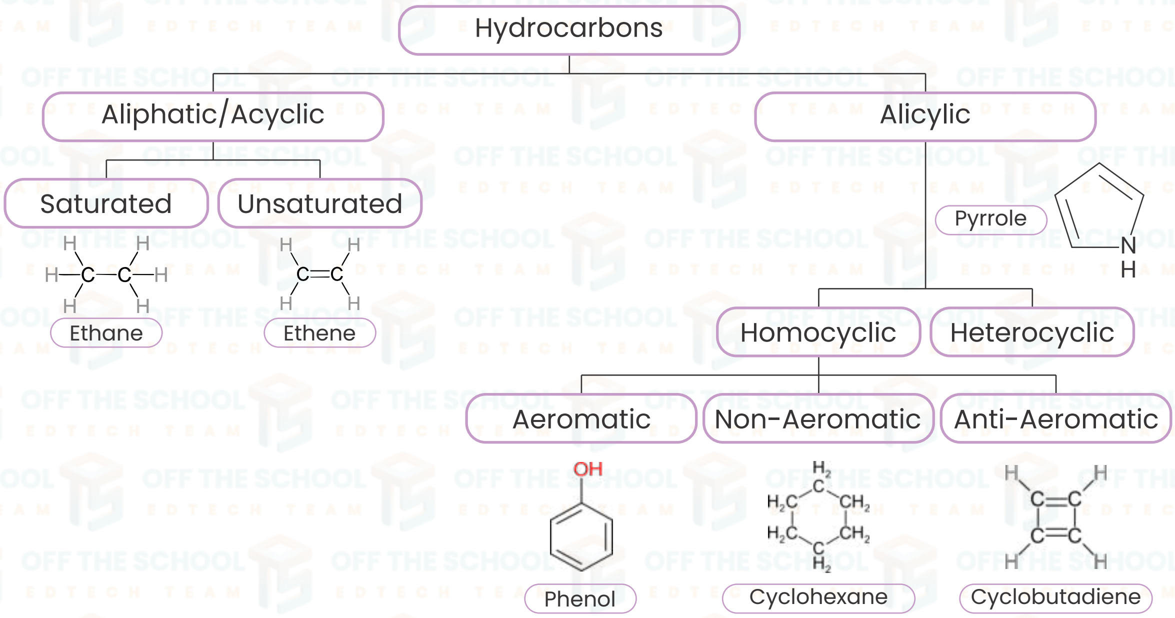
INTRODUCTION TO HYDROCARBON



HYDROCARBONS:

"Organic compounds which contain carbon & hydrogen only are called hydrocarbons."

Classification of hydrocarbons:

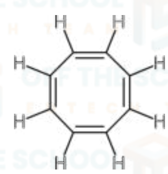


Aliphatic/Acyclic: These are hydrocarbons that do not have rings in their structure. They can be straight or branched chains.

- **Saturated:** These hydrocarbons have single bonds only. All the carbons are saturated with hydrogen. An example is ethane (C_2H_6), which has no double or triple bonds between carbon atoms.
- **Unsaturated:** These have one or more double or triple bonds between carbon atoms, which means they can hold more hydrogen if the double or triple bonds were to break. Ethene (C_2H_4), with a double bond, is an example.

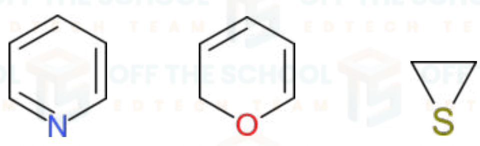
Alicyclic: These are hydrocarbons that have ring structures but are not only aromatic. They can be considered a type of aliphatic hydrocarbon.

- **Homocyclic:** These ring structures contain only carbon atoms in the ring. They can be either aromatic or non-aromatic.
- **Aromatic:** A special type of homocyclic compounds that are highly stable due to a continuous overlap of p-orbitals, following Hückel's rule.
- **Non-Aromatic:** These are cyclic compounds that do not follow Hückel's rule for stability. They do not have the special stability of aromatic compounds. Cyclo-butadiene is an example.
- **Anti-Aromatic:** These are also cyclic and planar like aromatic compounds, but instead of having a stabilizing number of electrons (4n+2), they have an unstable number of electrons (4n), making them less stable., Cyclo-octatetraene shown as a tub-shaped molecule, is an example.



- **Heterocyclic:** These are cyclic compounds that contains at least one atom other than carbon in the ring, such as sulfur, oxygen, or nitrogen.

For e.g.,

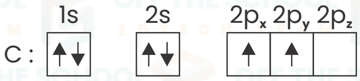


Pyridine Pyran

Structure Of Alkane (Sp^3 Hybridization in ethane)

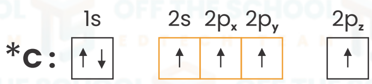
Hybridization of Carbon

Ground State

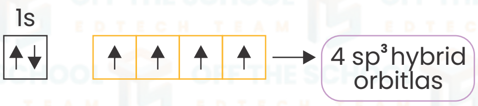


Total covalent bonds	07
Total sigma bonds	07
Total pi-bonds	00
Type of σ bond	Sp^3 - s: 06, Sp^3 – Sp^3: 01
Bond angle	109.5°
Bond length	1.54°
Shape	Tetrahedral

Excited State



Hybridized State



Sp^3 hybridization: is the type of hybridization in which one s and three p orbitals overlap each other to form four sp^3 hybridized orbitals.

Sp^3 hybridization in ethane:

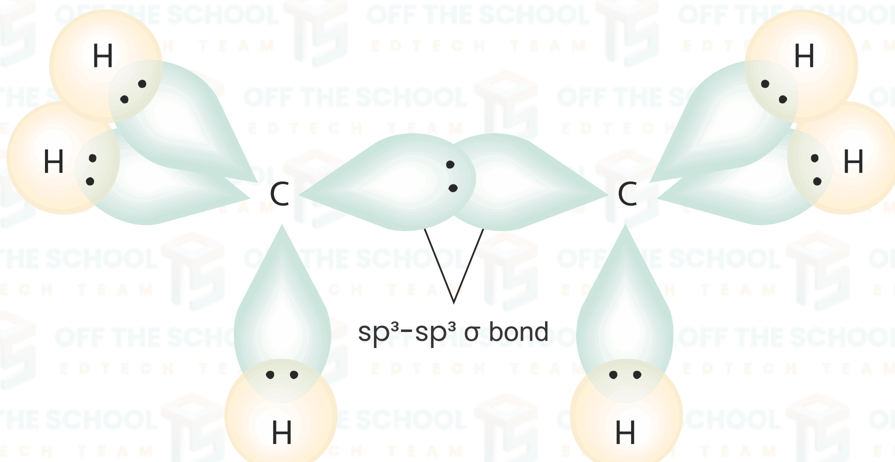
• Ethane is composed of two carbon atoms and six hydrogen atoms, giving it the molecular formula C_2H_6.

• In the formation of bonds, the 2s orbital and three 2p orbitals of each carbon atom mix together to form four sp^3 hybridized orbitals.

• The resulting geometry of ethane is tetrahedral, with bond angles of 109.5°.

• Each sp^3 hybrid orbital on the carbon atom overlaps with an s orbital of a hydrogen atom, and the sp^3 hybrid orbital of the other carbon atom to form six sigma (σ) bonds in total.

• One sigma (σ) bond is formed by the overlapping of Sp^3-Sp^3 hybrid orbitals of each carbon atom.



Homolytic Cleavage:

• **Definition:** Homolytic cleavage, also known as homolysis, occurs when a bond between two atoms breaks and each atom retains one of the shared electrons from the bond.

• **Example:** When a chlorine molecule (Cl_2) is exposed to ultraviolet light, the bond between the chlorine atoms breaks homolytically, resulting in two chlorine radicals:

$$Cl_2 \longrightarrow 2Cl\bullet$$

Heterolytic Cleavage:

• **Definition:** Heterolytic cleavage, or heterolysis, happens when a bond breaks and both of the shared electrons are taken by one of the atoms.

$$A:B \longrightarrow A^{+} + :B^{-}$$

FREE RADICAL SUBSTITUTION IN METHANE OR HALOGENATION IN METHANE:

Let's take the example of the chlorination of methane, where chlorine atoms replace hydrogen atoms in methane (CH_4), to produce chloromethane (CH_3Cl) and other substituted products like dichloromethane (CH_2Cl_2), chloroform ($CHCl_3$), and carbon tetrachloride (CCl_4). The process involves several steps:

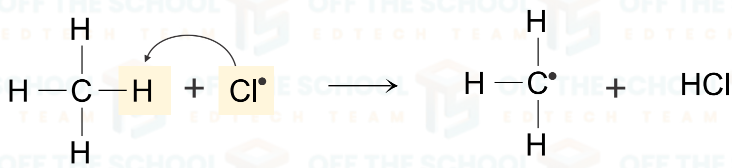
1. Initiation (Photochemical dissociation):

• Chlorine molecules (Cl_2) are broken down into chlorine radicals ($Cl•$) by heat or light.

$Cl_2 \longrightarrow 2Cl•$ (chlorine molecules break into chlorine radicals)

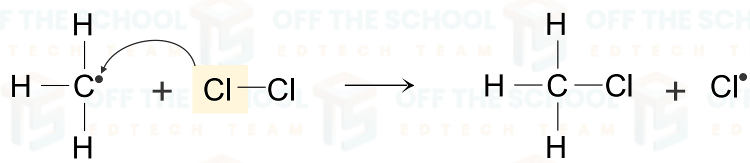
2. Propagation or elongation of chain:

In the 1st propagation step a chlorine radical reacts with a methane molecule, taking a hydrogen atom and forming hydrogen chloride (HCl) and a methyl free radical ($CH_3•$).



methane chlorine methyl free radical

In the 2nd propagation step the methyl free radical reacts with another chlorine molecule to form chloromethane and another chlorine radical, which can continue the chain.

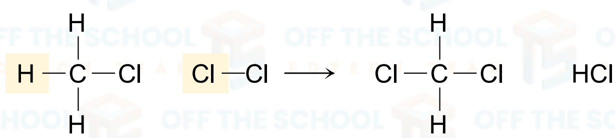


methyl free radical chloromethane (methyl chloride)

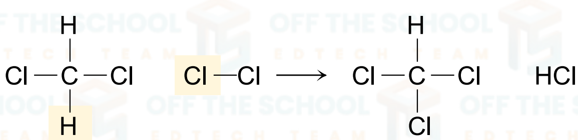
Note that only one half of the chlorine atoms occur in the organic product. The other half form hydrogen chloride, a commercially valuable compound.

3. Termination:

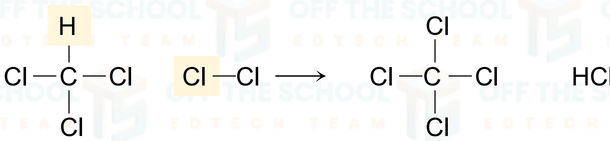
• Radicals react with each other to form stable molecules, thus ending the reaction. $Cl• + Cl• \longrightarrow Cl_2$ (two chlorine radicals recombine)



dichloromethane (methylene chloride)



trichloromethane (chloroform)



tetrachloromethane (carbon tetrachloride)

Structure Of Alkene (Sp^2 Hybridization in ethene)

Electronic configuration of C-atom

Ground State

C : 1s [↑↓] 2s [↑↓] $2p_x$ $2p_y$ $2p_z$ [↑][↓][ ]

Excited State

*C : 1s [↑↓] 2s $2p_x$ $2p_y$ [↑][↑][↑] $2p_z$ [↑]

Hybridized State

1s [↑↓] 3 sp² hybrid orbitlas [↑][↑][↑] $2p_z$ [↑]

Total covalent bonds:	06
Total sigma bonds	05
Total pi-bonds;	01
Type of σ bond:	Sp^2 - s: 04, Sp^2 – sp^2: 01
Type of π bond:	Pz-Pz
Bond angle:	120°
Bond length:	1.34°
Shape:	Trigonal planar

Sp² Hybridization: sp² hybridization is the type of hybridization in which one s orbital and two p orbitals overlap each other to form three sp² hybridized orbitals.

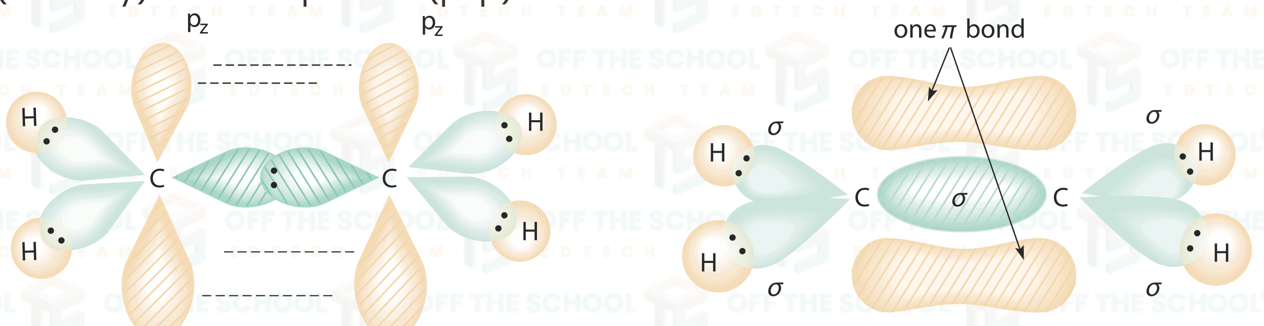
Sp² Hybridization in Ethene:

• Ethene consists of two carbon atoms and four hydrogen atoms; therefore, the molecular formula of ethene is C_2H_4.

• Before the formation of bonds, one 2s orbital and two 2p orbitals of each carbon atom are mixed together to form three sp² hybrid orbitals, while the remaining 2pz orbital of each carbon atom remains unhybridized and lies perpendicular to the plane of the sp² orbitals.

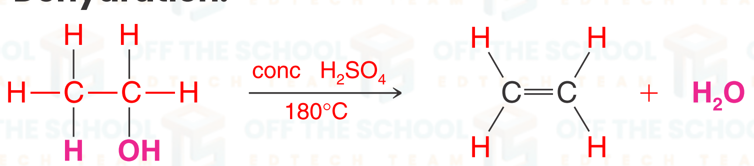
• Two sp² hybrid orbitals of each carbon atom overlap with the s orbital of two hydrogen atoms forming two sigma bonds (sp²-s), while the third sp² hybrid orbital overlaps with the sp² orbital on the other carbon atom forming another sigma bond (sp²-sp²). Therefore, there are overall four sigma bonds formed in each molecule.

• The unhybridized 2pz orbitals of each carbon atom are parallel to each other and perpendicular to the plane of the sp² hybrid orbitals. They overlap side by side (laterally) to form a pi bond (p-p).



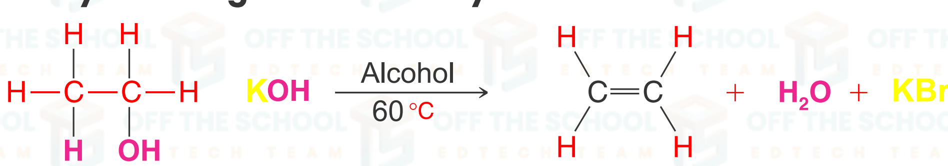
(a) (b)

Dehydration:



Ethanol

Dehydrohalogenation of Alkyl Halide:



Ethanol

Markovnikov's Rule:

It states that; when a protic acid (like hydrogen bromide, HBr) is added to an unsymmetrical alkene, the hydrogen atom from the acid will attach to the carbon with the greater number of hydrogen atoms, and the halide (or other group) will attach to the carbon with the fewer hydrogen atoms.

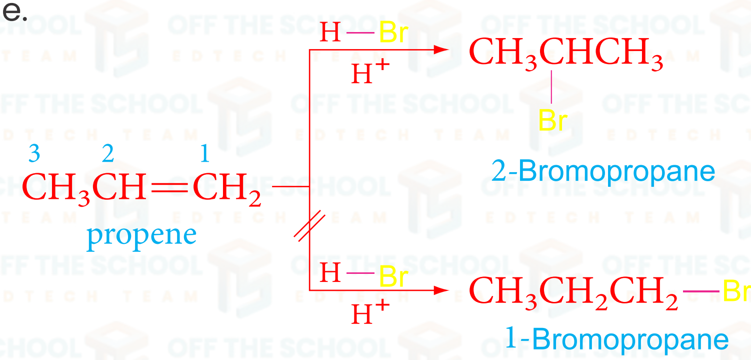
1. Propene is an asymmetrical alkene – the two carbons in the double bond have different numbers of hydrogen atoms.

2. When HBr is added, the hydrogen (H) from HBr will attach to the carbon with the greater number of hydrogen atoms.

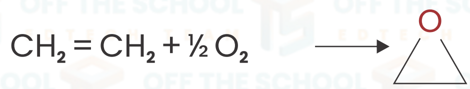
3. In propene, the carbon on the left (of the double bond) has two hydrogen atoms, while the carbon on the right has only one.

4. According to Markovnikov's rule, the hydrogen from HBr will attach to the left carbon, creating a more stable secondary carbocation.

5. The bromide (Br) ion will then attach to the carbon on the right, resulting in 2-bromopropane.

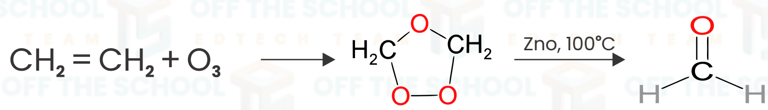


Epoxidation: An epoxide is a three-membered cyclic ether featuring an oxygen atom connected to two carbon atoms. Epoxidation of ethene involves the addition of an oxygen atom across the carbon-carbon double bond to form ethylene oxide.



Ozonolysis:

Ozonolysis is a reaction where ozone (O3) cleaves the double bond in ethene, forming an intermediate ozonide that subsequently decomposes into two aldehyde or ketone molecules. In the case of ethene, the reaction specifically yields two molecules of formaldehyde.

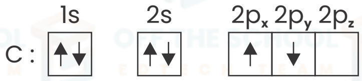


Sp Hybridization: sp hybridization is the type of hybridization in which one s orbital and one p orbital overlap each other to form two sp hybrid orbitals.

Structure of Alkyne: (sp Hybridization in Ethyne)

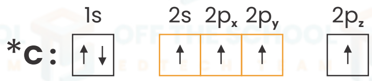
Electronic configuration of C-atom

Ground State

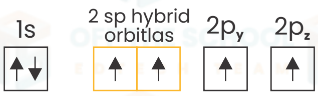


Total covalent bonds:	05
Total sigma bonds:	03
Total pi-bonds:	02
Type of σ bond:	Sp - s: 02, Sp – sp: 01
Type of π bond:	Py-Py & Pz-Pz
Bond angle:	180°
Bond length:	1.20°
Shape:	Linear

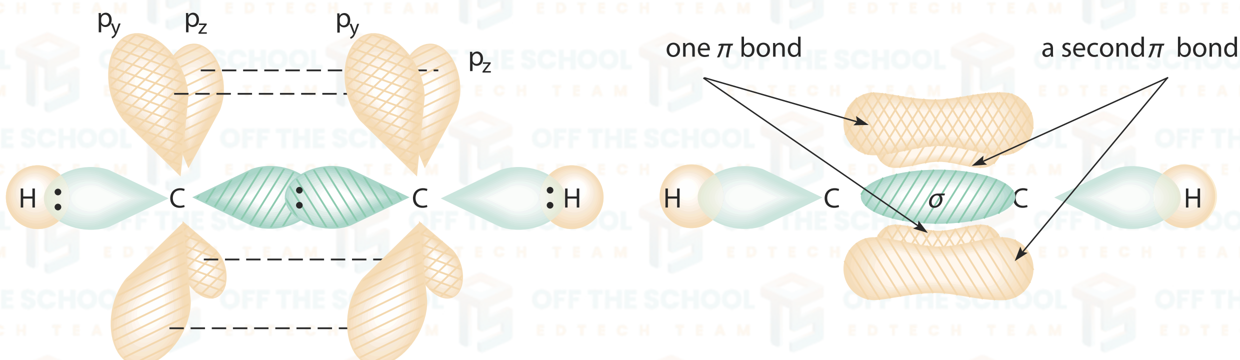
Excited State



Hybridized State



• Ethyne consists of two carbon atoms and two hydrogen atoms; therefore, the molecular formula of ethyne is C_2H_2.
• Before the formation of bonds, one 2s orbital and one 2p orbital of each carbon atom are mixed together to form two sp hybrid orbitals, while the remaining 2py and 2pz orbitals of each carbon atom stay unhybridized and lie perpendicular to the plane of the sp orbitals.
• One sp hybrid orbital of each carbon atom overlaps with the s orbital of a hydrogen atom forming one sigma bond (sp-s), while the other sp orbital overlaps with the sp orbital on the other carbon atom forming another sigma bond (sp-sp). Therefore, there are two sigma bonds formed in the molecule.
• The 2py and 2pz unhybridized orbitals of each carbon atom are parallel to each other and perpendicular to the line of the sp orbitals. They overlap side by side(laterally) to form two pi bonds (p-p).
• The geometry of ethyne is linear, and the bond angle between the hydrogen carbon bonds is 180°.



(a) (b)

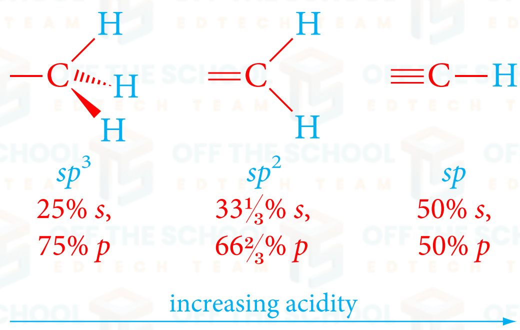
DISTINGUISH BETWEEN ALKYNE & ALKENE & VICE VERSA:

ACETYLIDE: When terminal hydrogen atom of alkyne is replaced with metal cation then it is known as acetylide or alkynides.

General Representation:

$$HC\equiv C-M^{+}$$
$$M^{+}-C\equiv C-M^{+}$$

ACIDITY OF TERMINAL ALKYNE: In ethyne and other terminal alkyne like propyne the hydrogen atom is bonded to the carbon atom with sp-sp overlap. As sp hybrid orbital has 50%'s' character in it and renders the carbon atom more electronegative. As a result, the sp hybrid carbon atom of a terminal alkyne pulls the electrons more strongly making them attached hydrogen atom slightly acidic and can be removed when treated with a mild base.



How can you distinguish between ethene and ethyne by simple chemical test?

To distinguish between ethene and ethyne, the compound in question should be reacted with ammoniacal silver nitrate. If a white precipitate forms, the compound is ethyne; if there is no reaction, the compound is ethene.

$$HC\equiv CH + 2AgNO_3 \xrightarrow{NH_3} \underline{Ag\text{-}C\equiv C\text{-}Ag} + 2HNO_3$$

Lunar caustic ammoniacal silver nitrate — White ppt of Silver acetylide/ethynide

$$HC\equiv CH + Cu_2Cl_2 \xrightarrow{NH_3} \underline{Cu-C\equiv C-Cu} + 2HCl$$

Reddish brown ppt
Copper acetylides or Copper ethynide.

Addition of Hydrogen Bromide:

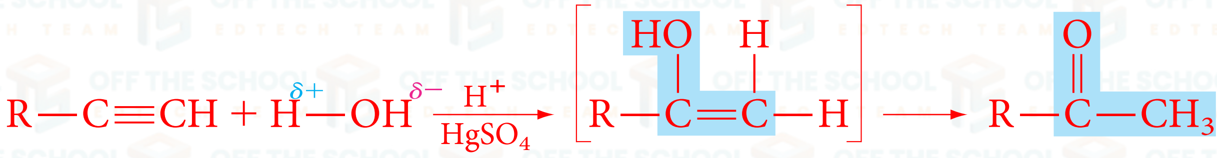
$$CH_3C\equiv CH + H-Br \longrightarrow CH_3\overset{+}{C}=CH_2 + Br^- \longrightarrow CH_3\underset{}{\overset{Br}{C}}=CH_2$$

2-bromopropene

$$CH_3\overset{Br}{C}=CH_2 + H-Br \longrightarrow CH_3\underset{Br}{\overset{+}{C}}-CH_3 + Br^- \longrightarrow CH_3-\underset{Br}{\overset{Br}{C}}-CH_3$$

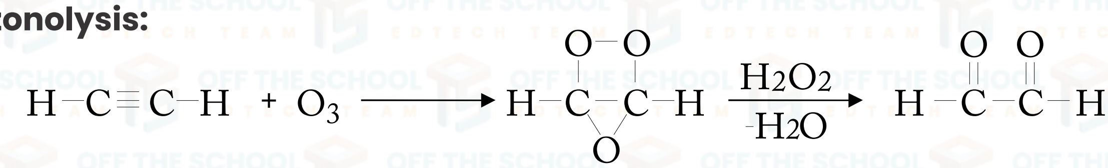
2,2-dibromopropane

Oxymercuration or Hydration of Ethyne:



a vinyl alcohol, or enol   a methyl ketone

Ozonolysis:



(ethyne)   (ozonide)   (glyoxal)

Isomerism:

Two or more compounds with the same molecular formula but different structural or physical and chemical properties are known as isomers, and this property of a compound is known as Isomerism.

Types of Isomerism:

Isomerism can be classified into two categories as follows:

✓ Structural or constitutional isomerism
✓ Stereoisomerism

Structural isomerism: is when molecules have the same number and types of atoms but are put together in different ways. It's like having the same set of building blocks but building different shapes each time. These different shapes are called structural or constitutional isomers.

The following are some examples of structural isomerism in various forms:

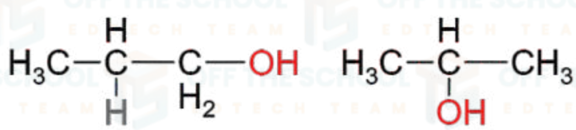
1. Chain isomerism
2. Position isomerism
3. Functional isomerism
4. Metamerism

1. Chain Isomerism: The type of structural isomer in which; molecules have the same atoms but form different shapes of carbon chains, like a straight line or a branch.

But-1-ane   ***2-Methylprop-1-ane***   ***But-2-ene***   ***2-Methylprop-1-ene***

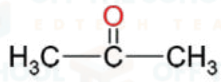
Alkane   **Alkene**

2. Position Isomerism: This happens when the same atoms are connected in the same order, but functional group, like a double bond or a chlorine atom, is in a different position on the chain.

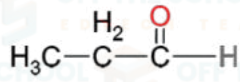


Prop-1-ol   ***Prop-2-ol***

Functional Isomerism: Functional group isomers are molecules that share an identical molecular formula but are distinct from one another due to the variation in the type of functional group present.



Propanone

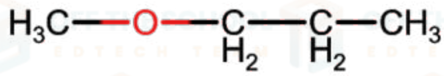


Propanal

Metamerism: Metamerism occurs in organic chemistry when an organic compound displays a difference in the arrangement of alkyl groups on either side of its functional group, despite having the same molecular formula.

1-Methoxyethane

1-Ethoxyethane



1-Methoxypropane

Stereoisomerism:

Stereoisomers are compounds that have the same chemical and structural formulae but differ in the relative or spatial or three-dimensional arrangement of the atoms or groups in space, a phenomenon known as stereoisomerism.

Geometrical Isomerism:

Isomerism generated by constrained rotation around a link in a molecule is known as geometrical and also called cis-trans isomerism. A wide range of substances exhibits geometrical isomerism, which can be characterized as follows:

Conditions for geometrical isomerism:

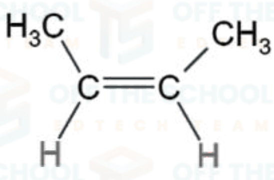
All alkenes don't show geometrical isomerism. Necessary conditions are:

✓ There must be double bond (=) present between two carbon atoms.

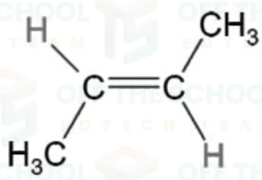
✓ The groups attached with single carbon atom must be different.

Cis-isomer: is one in which two similar groups are on the same side of double bond.

Trans-isomer: is one in which two similar groups are on the opposite side of the double bond.

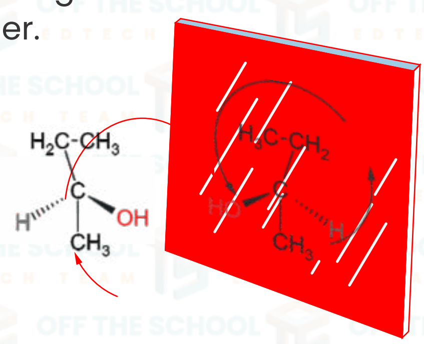


***cis*-but-2-ene**



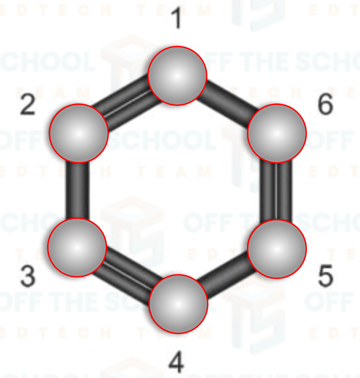
***trans*-but-2-ene**

Optical isomers: Optical isomers are two compounds that contain the same number and type of atoms, bonds, and spatial configurations of the atoms but are not superimposable mirror images of each other.

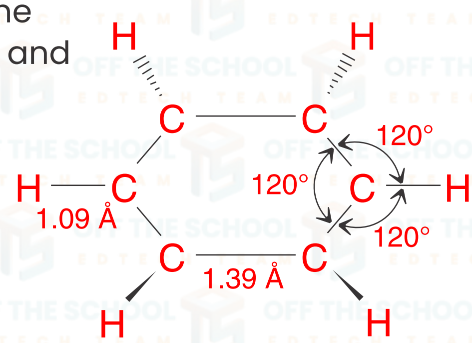


Chiral carbon: The carbon atom, which is attached to four different functional groups or molecules, is known as chiral carbon.

CHEMISTRY OF BENZENE



- ✓ **Chemical Name:** Benzene.
- ✓ **Molecular Formula & mass:** C_6H_6 & 78 a.m.u.
- ✓ **Class of compounds:** Aromatic Hydrocarbons.
- ✓ **Occurrence:** Coal Tar and Petroleum.
- ✓ **Hybridization:** sp^2.
- ✓ **Geometry:** Planar hexagonal cyclic structure.
- ✓ **Bond Angle:** 120°.
- ✓ **Bond Length:** 1.39 Å.
- ✓ **Number of Sigma Bonds:** 12 (6 σ bonds between the carbon atoms and 6 σ bonds between each carbon and a hydrogen atom.)
- ✓ **Number of Pi Bonds:** 3.
- ✓ **Isolation from coal tar:** 1825 by M. Faraday.
- ✓ **Discovery:** 1825 by M. Faraday.
- ✓ **Discovery of Structure:** 1865 by A. Kekulé.



Molecular Orbital treatment of Benzene:

✓ In benzene each C-atom is in a state of sp^2 hybridization because each C-atom is attached to three atoms. Therefore, it forms three sp^2 hybridized orbitals.

Ground State

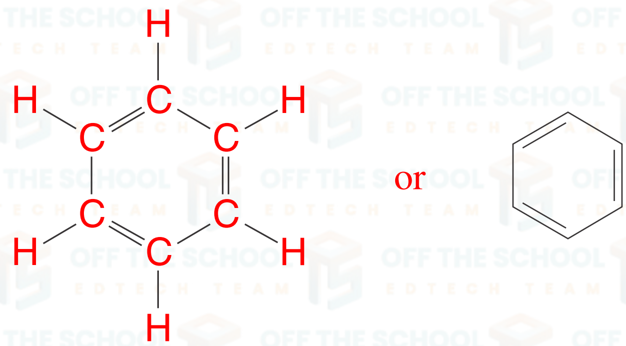
	1s	2s	$2p_x$	$2p_y$	$2p_z$
C :	↑↓	↑↓	↑	↓	

Excited State

	1s	2s	$2p_x$	$2p_y$	$2p_z$
*C :	↑↓	↑	↑	↑	↑

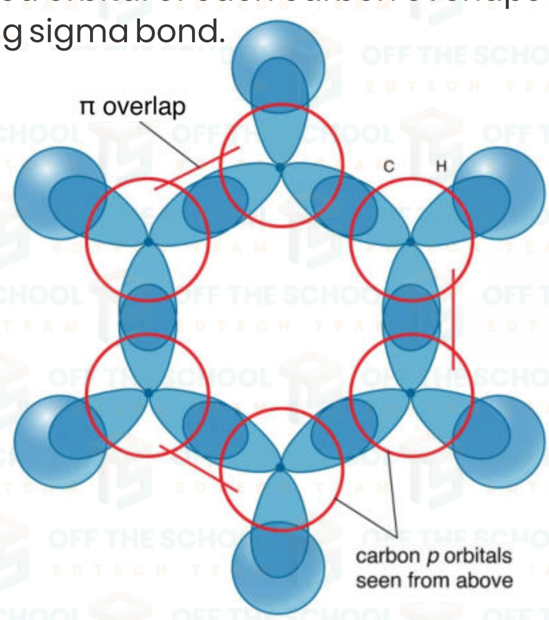
Hybridized State

1s	3 sp² hybrid orbitlas			$2p_z$
↑↓	↑	↑	↑	↑



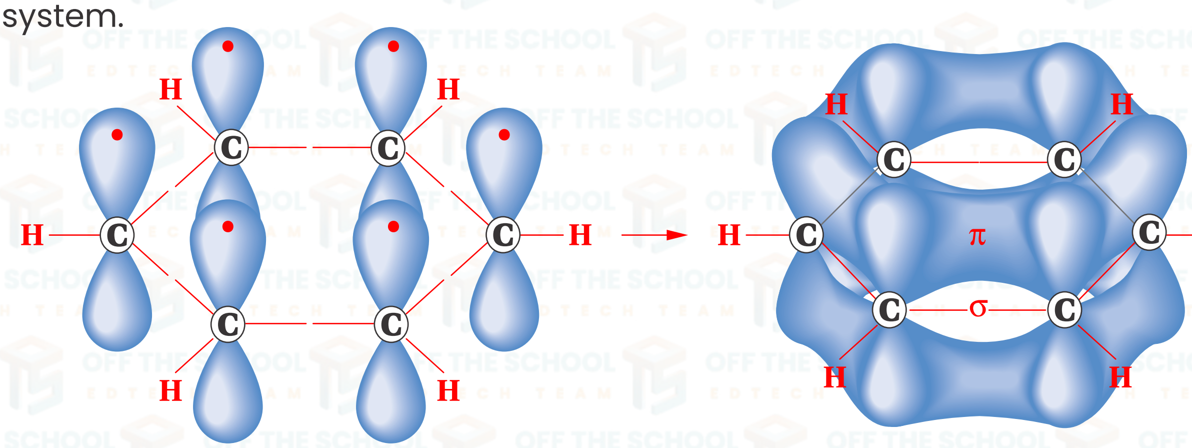
Kekulé structure for benzene **Bond-line representation of Kekulé structure**

Formation of sigma bonds: Two sp^2 hybridized orbitals overlap with two sp^2 hybridized orbitals of two other carbon atoms to form two sigma bonds. Similarly one sp^2 hybridized orbital of each carbon overlaps with 1s atomic orbital of hydrogen atom forming sigma bond.

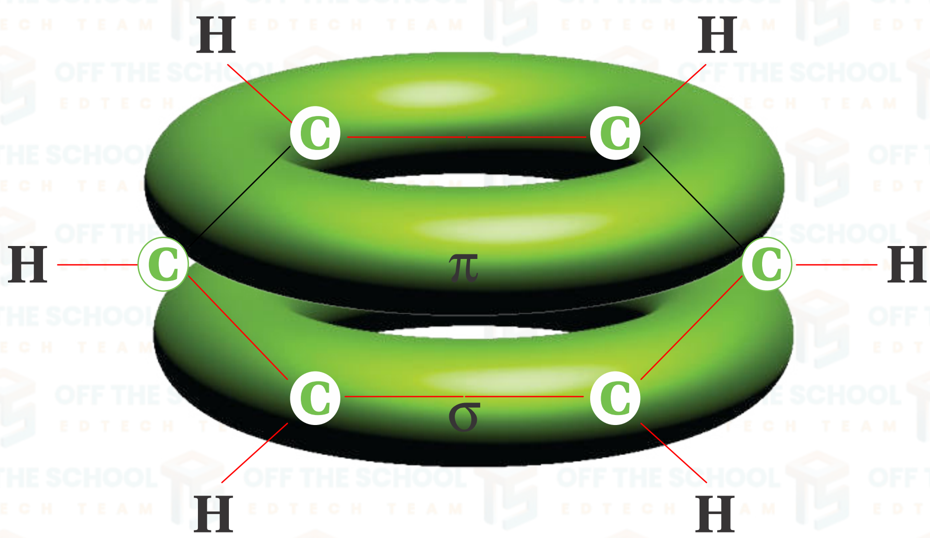


P-Orbitals:
There are six unhybridized p-orbitals, one on each carbon atom. These orbitals are positioned perpendicular to the sigma bonds. Each p-orbital overlaps with the p-orbitals of neighboring carbon atoms in a parallel way, forming a continuous cloud of negative charge.

Pi-Bonding: It results in an extensive delocalized electron cloud and a pi-bonding system.



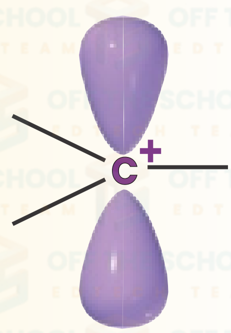
Stability: Delocalization of p-orbitals over the entire ring produces sandwich like structure of benzene and decreases the energy of molecule. Consequently, the molecule becomes more stable and less reactive.



A **carbocation** is a general name for a species with a **positive** charge at **carbon.**

Formal charge of +1 on carbon

- Empty p-orbital
- Trigonal planar geometry
- Lewis acidic (i.e., accepts a pair of electrons)
- 6 valence electrons
- sp^2 hybridization



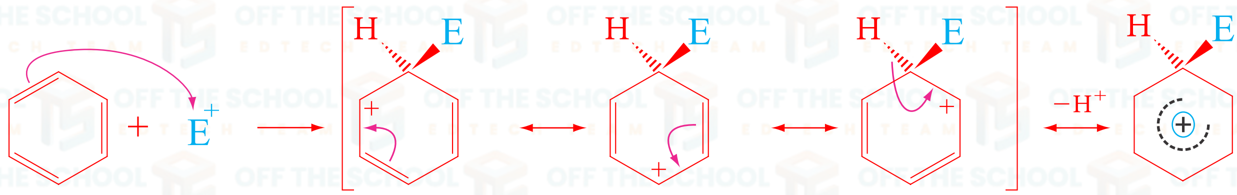
ELECTROPHILIC AROMATIC SUBSTITUTION (EAS):

Definition: When an electrophile attacks a benzene ring, replacing one of the hydrogen atoms with itself, the reaction is called electrophilic aromatic substitution.

Explanation: Electrophilic aromatic substitution (EAS) is a fundamental type of reaction in organic chemistry, whereby an electrophile selectively replaces a hydrogen atom in an aromatic ring such as benzene. This process retains the aromaticity of the benzene ring. Typical electrophiles involved in such reactions include halogens (X), nitronium ions (NO_2^+), and sulfonium ions (SO_3^+).

Mechanism: The EAS mechanism generally involves two main steps:

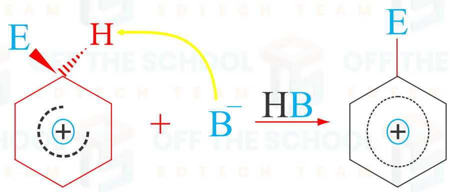
1. The electrophile attacks the π-electron system of the benzene, forming a non aromatic carbocation intermediate known as the sigma complex or arenium ion.



benzene

sigma complex or arenium or benzonium ion

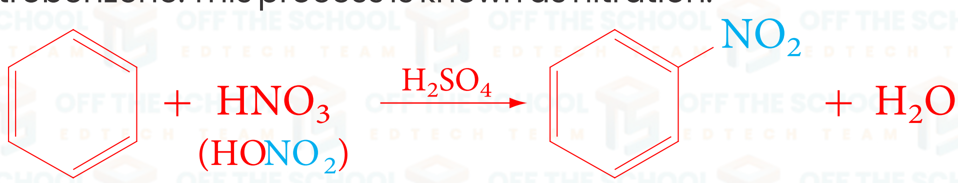
2. A base, often the solvent or a by-product of the electrophile generation, removes a proton from the sigma complex, restoring aromaticity to the ring.



Substituted Product

Nitration:

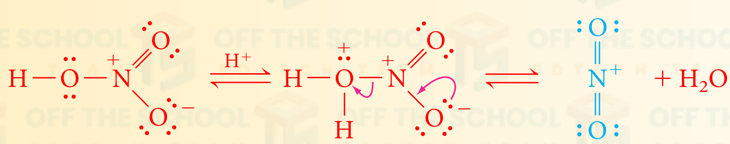
Definition: When benzene is treated with a nitrating mixture, which is a combination of concentrated nitric acid and concentrated sulfuric acid usually in varying proportions depending on the specific conditions, a hydrogen atom on the benzene ring is replaced by a nitro group ($-NO_2$). This results in the formation of nitrobenzene. This process is known as nitration.



nitration

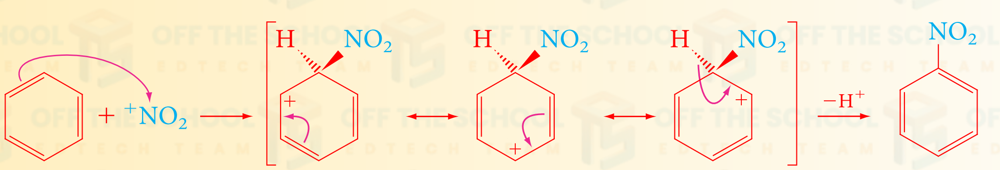
Nitration

In aromatic nitration reactions, the sulfuric acid catalyst* protonates the nitric acid, which then loses water to generate the **nitronium ion** (NO_2^+), which contains a positively charged nitrogen atom.



nitric acid — protonated nitric acid — nitronium ion

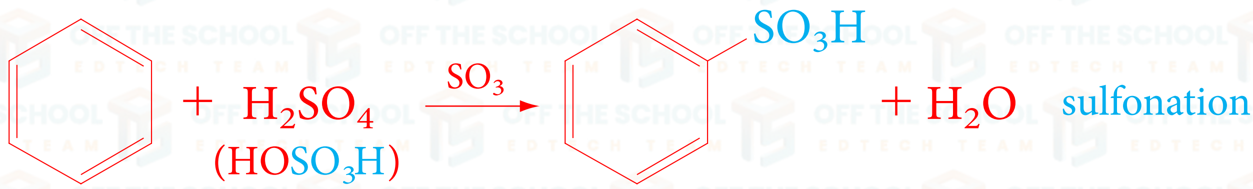
The nitronium ion, a strong electrophile, is then attacked by the aromatic ring.



benzene — nitrobenzene

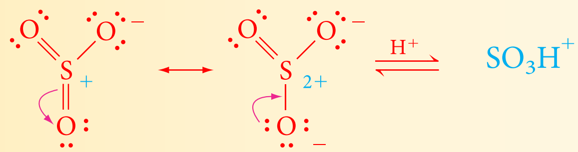
Sulphonation:

Definition: Sulfonation of benzene is a reaction where benzene is treated with sulfur trioxide (SO_3) or fuming sulfuric acid, also known as oleum, to replace one of the hydrogen atoms on the benzene ring with a sulfonic acid group ($-SO_3H$). This process results in the formation of benzenesulfonic acid and is known as sulfonation.

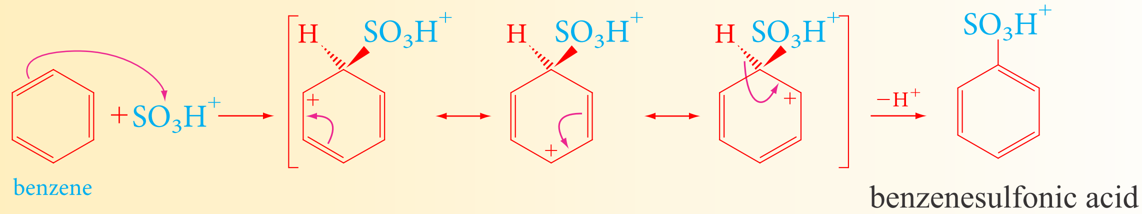


Sulfonation

In aromatic sulfonation reactions, we use either concentrated or fuming sulfuric acid, and the electrophile may be sulfur trioxide, SO_3, or protonated sulfur trioxide, $^{+}SO_3H$. The following resonance structures demonstrate that SO_3 is a strong electrophile at sulfur.



The nitronium ion, a strong electrophile, is then attacked by the aromatic ring.



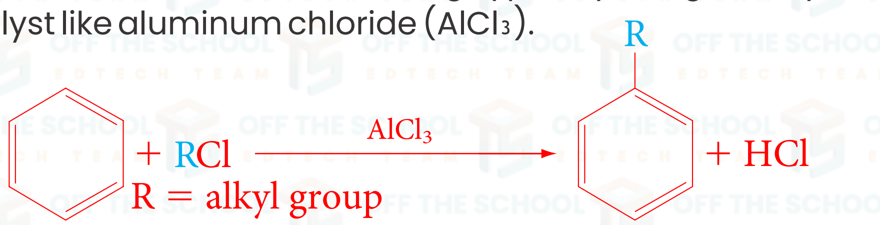
*Sulfuric acid provides a proton catalyst as shown in the following equilibrium:

$H{-}OSO_3H \rightleftharpoons H^{+} + {}^{-}OSO_3H.$

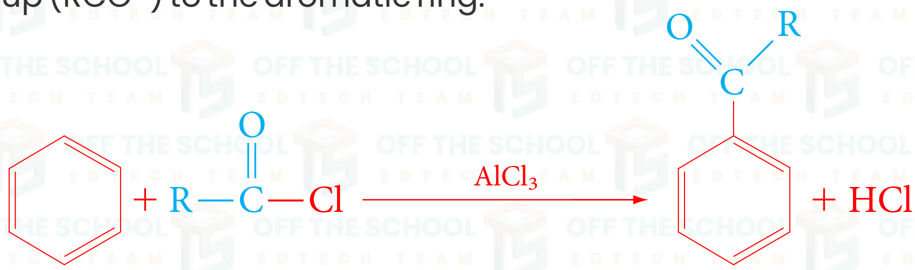
Friedel Crafts Alkylation and Acylation:

Definition: Friedel-Crafts reactions are a type of chemical reaction that allow an aromatic ring, like benzene, to form new carbon-carbon (C-C) bonds. This happens when the aromatic ring is treated with either alkyl or acyl halides in the presence of a strong Lewis acid, such as aluminum chloride ($AlCl_3$).

Alkylation: Friedel-Crafts alkylation is a chemical reaction in which an alkyl group is introduced to an aromatic ring, typically using an alkyl halide and a Lewis acid catalyst like aluminum chloride ($AlCl_3$).

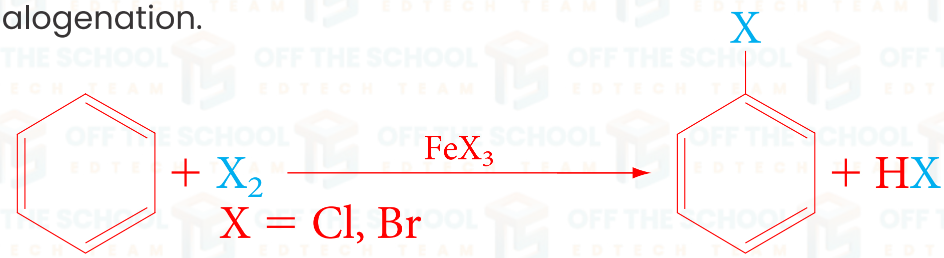


Acylation: Friedel-Crafts acylation uses acyl halides and results in the addition of an acyl group (RCO-) to the aromatic ring.



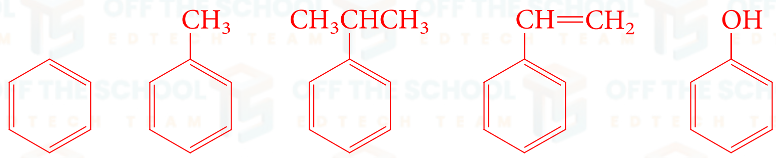
Halogenation:

Definition: When a benzene ring, or any aromatic compound, is treated with a halogen (like chlorine or bromine) in the presence of a Lewis acid catalyst (such as iron tribromide, $FeBr_3$, for bromination or iron trichloride, $FeCl_3$, for chlorination), a hydrogen atom on the ring is replaced by a halogen atom. This reaction is called aromatic halogenation.

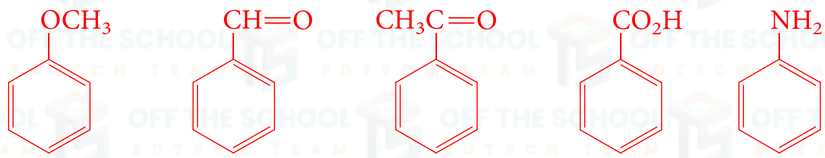


Nomenclature of Aromatic Compounds

Because aromatic chemistry developed in a haphazard fashion many years before systematic methods of nomenclature were developed, common names have acquired historic respectability and are accepted by IUPAC. Examples include:

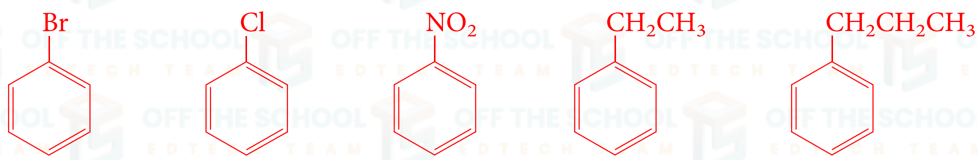


benzene, toluene, cumene, styrene, phenol



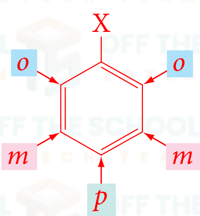
anisole, benzaldehyde, acetophenone, benzoic acid, aniline

Monosubstituted benzenes that do not have common names accepted by IUPAC are named as derivatives of benzene.

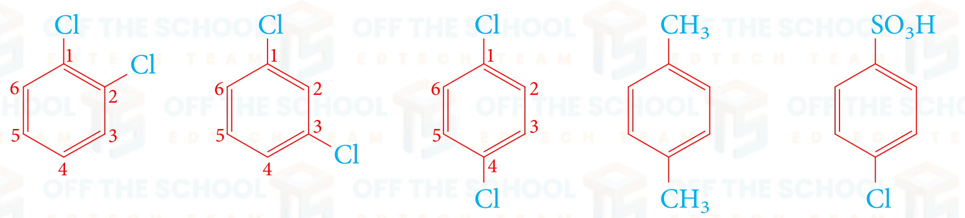


bromobenzene, chlorobenzene, nitrobenzene, ethylbenzene, propylbenzene

When two substituents are present, three isomeric structures are possible. They are designated by the prefixes ***ortho***-, ***meta***-, and ***para***-, which are usually abbreviated as ***o***-, ***m***-, and ***p***-, respectively. If substituent X is attached (by convention) to carbon 1, then *o*-groups are on carbons 2 and 6, *m*-groups are on carbons 3 and 5, and *p*-groups are on carbon 4.*

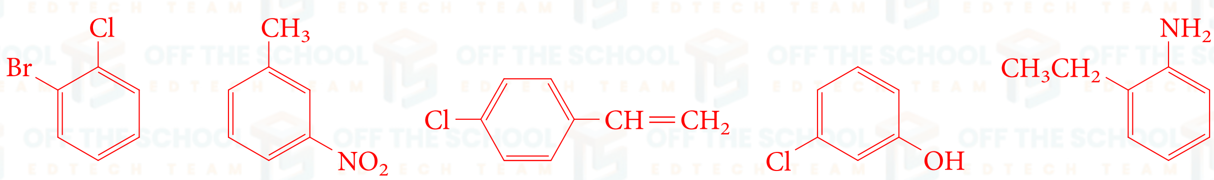


Specific examples are



ortho-dichlorobenzene, *meta*-dichlorobenzene, *para*-dichlorobenzene, *para*-xylene**, *para*-chlorobenzenesulfonic acid

The prefixes *ortho*-, *meta*-, and *para*- are used even when the two substituents are not identical.



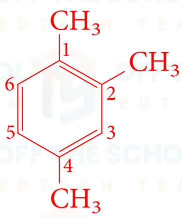
o-bromochlorobenzene (note alphabetical order), *m*-nitrotoluene, *p*-chlorostyrene, *m*-chlorophenol, *o*-ethylaniline

*Note that X can be on any carbon of the ring. It is the *position of the second substituent relative to that of X* that is important.

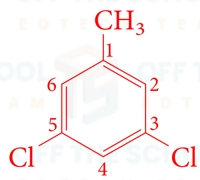
**The common and IUPAC name is xylene, *not p*-methyltoluene.

Although *o*-, *m*-, and *p*- designations are commonly used in naming disubstituted benzenes, position numbers of substituents can also be used. For example, *o*-dichlorobenzene can also be named 1,2-dichlorobenzene, and *m*-chlorophenol can also be named 3-chlorophenol.

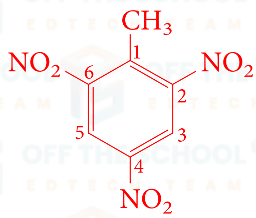
When more than two substituents are present, their positions are designated by numbering the ring.



1,2,4-trimethylbenzene



3,5-dichlorotoluene



2,4,6-trinitrotoluene (TNT)

PROBLEM

Draw the structure of

a. *o*-dinitrobenzene
b. *m*-nitrotoluene
c. *p*-bromophenol
d. 4-chloroaniline
e. *p*-vinylbenzaldehyde
f. 1,4-dichlorobenzene

Draw the structure of

a. 1,3,5-trimethylphenol
b. 4-ethyl-2,6-difluorotoluene